Pediatric Musculoskeletal Disorders

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Objectives

- Recognize and describe the following pediatric orthopedic problems:
  - Talipes Equinovarus
  - Metatarsus Adductus
  - Calcaneal Apophysitis
  - Femoral Antevrersion
  - Internal Tibial Torsion
  - Tibia Vara
  - Osgood-Schlatter Disease
  - Developmental Dysplasia of the Hip
  - Transient Synovitis
  - Legg-Calve-Perthes Disease
  - Slipped Capital Femoral Epiphysis
  - Scoliosis
  - Kyphosis
  - Torticollis
  - Spondylolysis
  - Spondylolisthesis
  - Diskitis
  - Radial Head Subluxation

- Identify the most common presentations; and describe the diagnostic tests and treatment for the above.

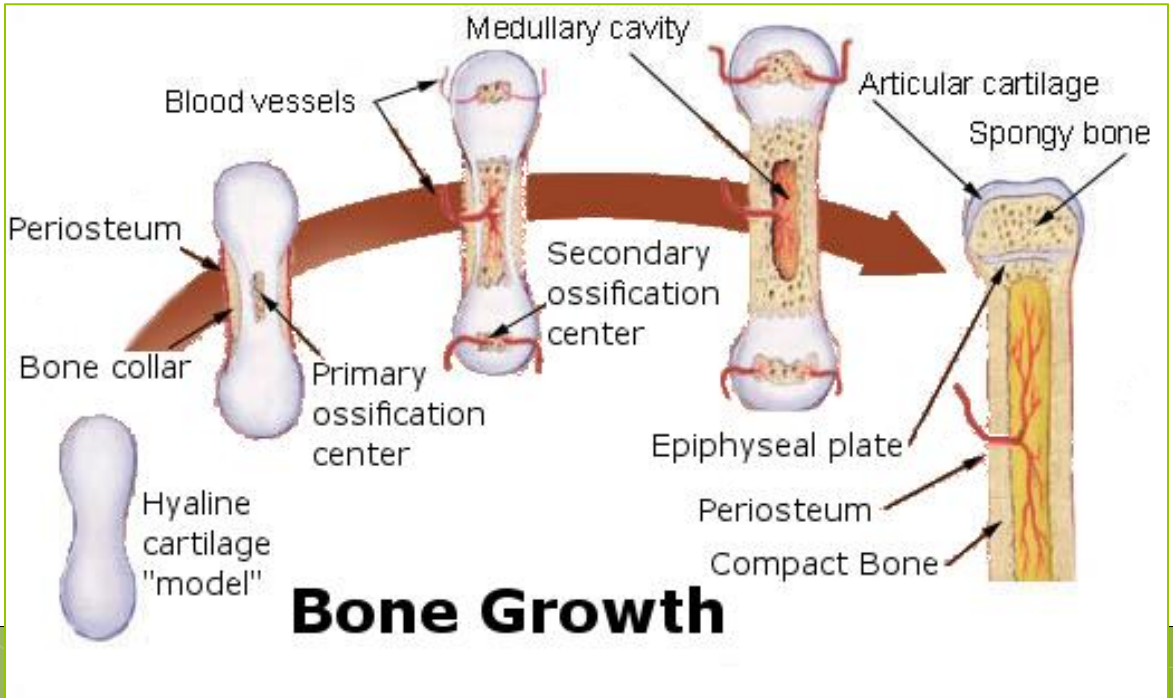
Recommended Reading

- Kliegman RM. *Nelson: Essenstials of Pediatrics*. 7th Edition. Saunders; 2014
  - Chapters 199-203
  - *Text available online via TUCOM library (Call# RJ45)*

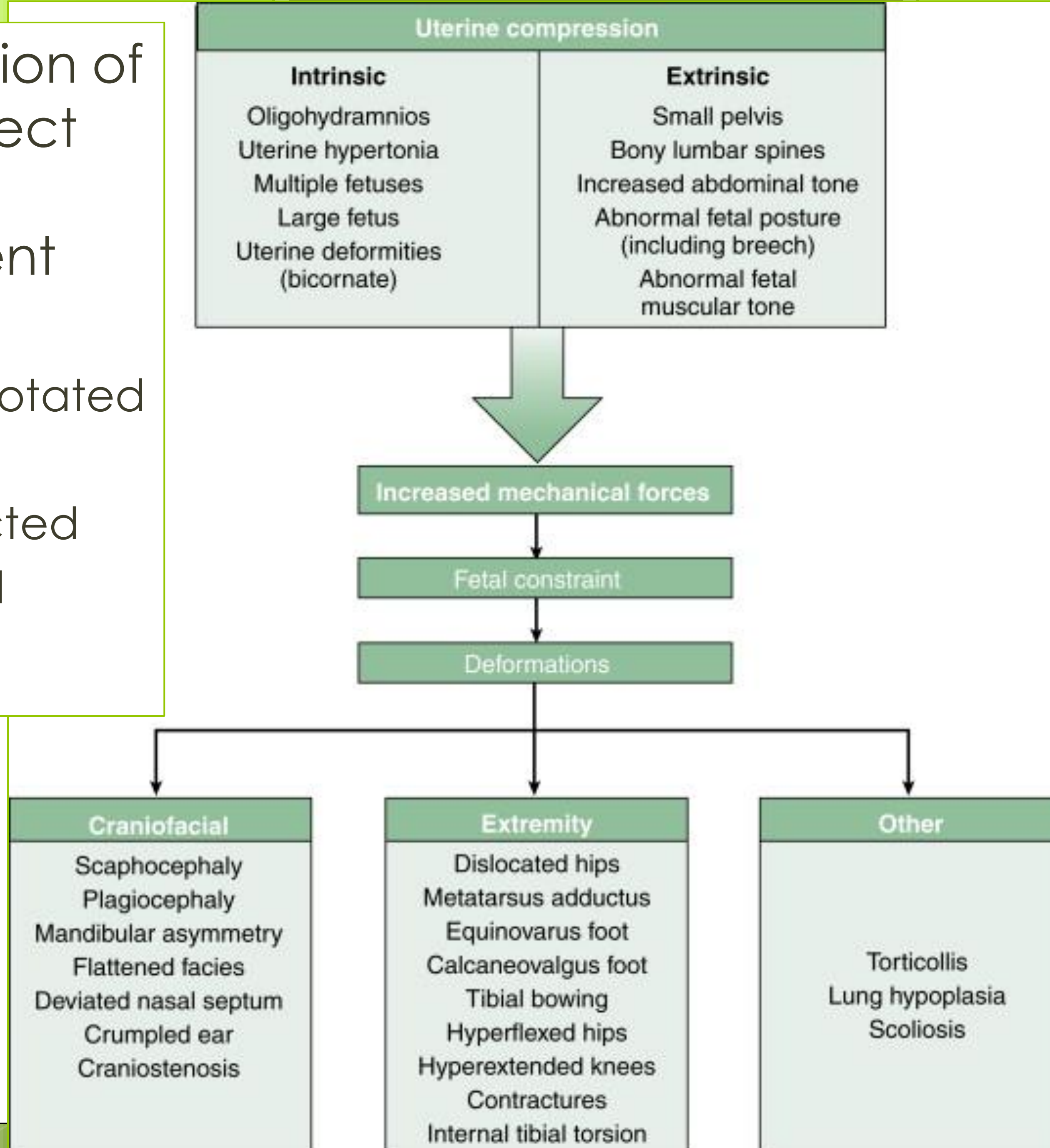
- ***Quizlet:*** **https://quizlet.com/61140828/pediatric-musculoskeletal-disorders-flash-cards/ = Pediatric Musculoskeletal Disorders**

Pediatric Orthopedic Considerations

- Growth and development
  - Ends of long bones contain a higher proportion of cartilage
    - Allows for a unique vulnerability to trauma and infection



- The in utero position of the fetus can affect the angular & torsional alignment
  - Newborns:
    - Hips externally rotated
    - Ankles inverted
    - Forefoot adducted
    - Born with flexed posture



The Foot

Talipes Equinovarus
Metatarsus Adductus
Polydactyly
Syndactyly
Calcaneal Apophysitis

The Foot

- For non-weight bearing:

Posturing	Habitual position in which infant holds foot	Passive ROM is NORMAL
Deformity	Appears similar to posturing	Passive ROM is RESTRICTED

Talipes Equinovarus (Club Foot)

- Involves **entire leg**
- 1 in 1000 newborns
- 50% bilateral
- Abnormal tarsal interactions →
  - Generalized limb hypoplasia →
    - Shortening of foot
- Congenital (75%), Teratologic (myelomeningocele, arthrogryposis), or Positional

Talipes Equinovarus (Club Foot)

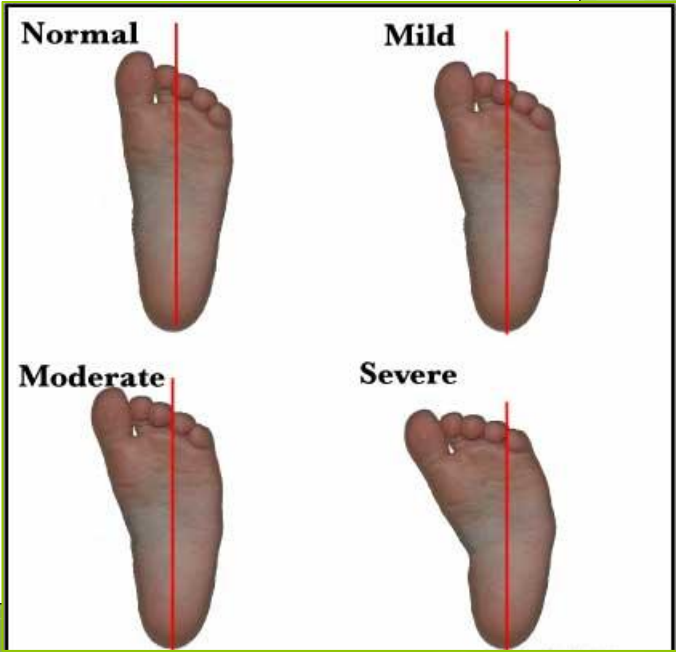
- **Hindfoot equinus** (heel pulled upward) and **varus**
- **Forefoot adduction**
- Varying degrees of rigidity
- Dx: X-rays not needed (navicular ossifies ~3-4 years)
- Tx: Serial casting; Achilles tenotomy

Hindfoot Varus (Sign of a high arched foot) = Heel Points to the midline


Clubfoot treatment over 4 – 6 weeks
Stage 1
Stage 2
Stage 3
Stage 4
Stage 5

Metatarsas Adductus

- **Most common** foot disorder in infants
- **Convexity of lateral foot**
- Caused by **in utero** positioning
  - More common in primigravid
- Bilateral 50%
- 2% also have DDH



Metatarsus Adductus

- **Dx: Clinical**
- **Tx: Resolves spontaneously** >90%
- If > 2 years old: serial casting, bracing, or surgery

Polydactyly & Syndactyly

- **Polydactyly (extra digits)**
- **Syndacyly (fusion of digits)**
  - More common

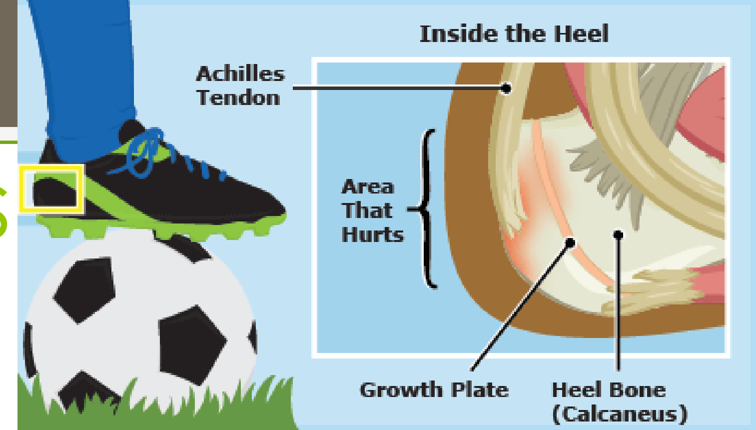
- Both usually **benign cosmetic** problems
- May be associated with syndromes

POLYDACTYLY	SYNDACTYLY
Ellis-van Creveld syndrome Rubinstein-Taybi syndrome Carpenter syndrome Meckel-Gruber syndrome Polysyndactyly Trisomy 13 Orofaciodigital syndrome	Apert syndrome Carpenter syndrome de Lange syndrome Holt-Oram syndrome Orofaciodigital synd Polysyndactyly Fetal hydantoin sync Laurence-Moon-Biec Fanconi pancytoper Trisomy 21 Trisomy 13 Trisomy 18

Calcaneal Apophysitis (Sever Disease)

- Common cause of **heel pain in active young people**
- Mean age = 9 years (♀), 11-12 years (♂)
- B/L 60%
- Calf muscle →
  - Achilles Tendon →
    - **Microfracture of calcaneal apophysis**

Calcaneal Apophysitis (Sever Disease)



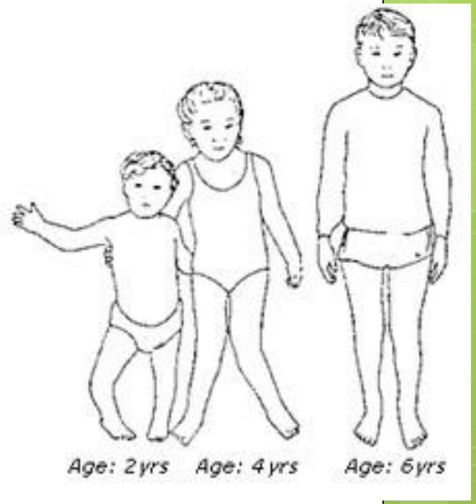
- HPI: **young athlete, heel pain with activity, pain decreases with rest**
- Rare swelling; +/- limp
- **Tenderness to palpation** of posterior calcaneus, hypertonic Achilles
- Tx: **Rest, ice, NSAIDs, stretching**, +/- heel wedges

The Lower Extremity & Knee

In-toeing
Femoral Anteversion
Internal Tibial Torsion
Out-toeing
Genu Valgum
Genu Varum
Tibia Vara
Osgood-Schlatter Disease

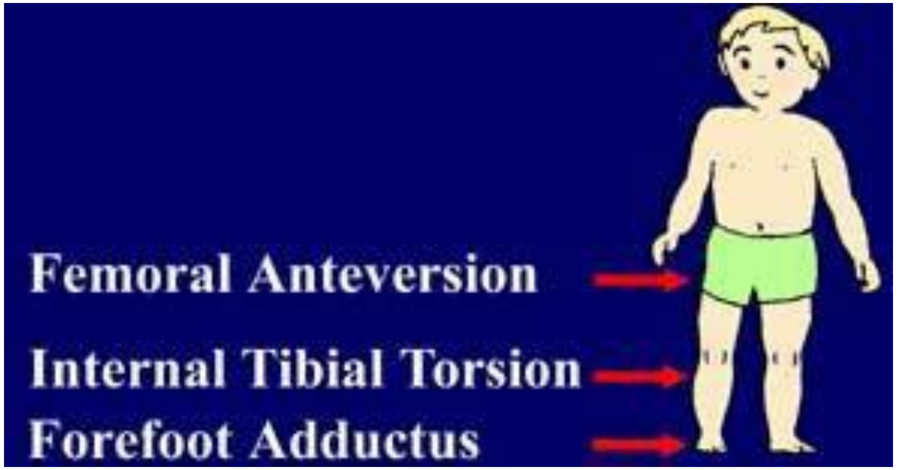
Parental Concern

- In-toeing, out-toeing, physiologic “bowlegs”, and “knock knees” are common reasons parents bring their child in
- Most of these are **physiologic**
  - Should **not** cause limp or pain
  - Bilateral
- Resolve with normal growth
- Femur & Tibia = internally rotated ~30 degrees at birth



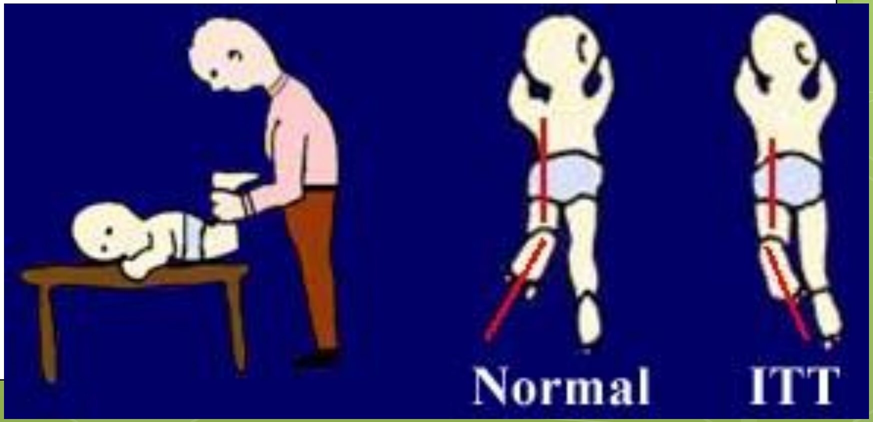
Most Common Causes of **In-Toeing**

- **Metarasus Adductus** = babies (<1 year)
- **Internal Tibial Torsion** = toddlers (1-2 years)
- **Femoral Anteversion** = school age kids (>2 years)



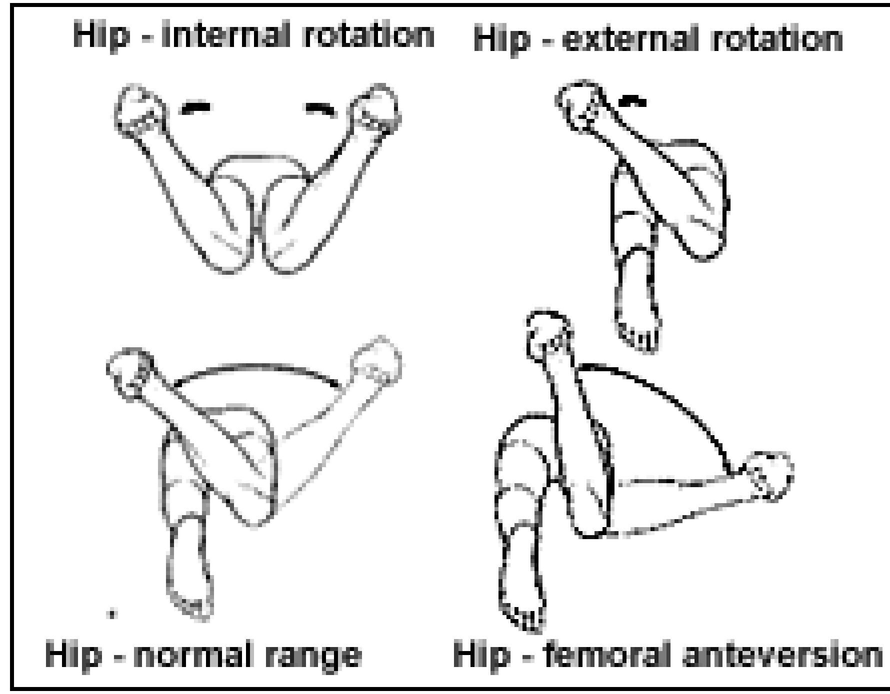
Internal Tibial Torsion

- HPI: **Toddler with in-toeing**
- PE: **Measure thigh-foot angle**
  - Patient lies prone with knee flexed to 90 degrees
- Tx: **Reassurance**, follow-up
  - Resolves by 7-8 years
  - <1% require surgery



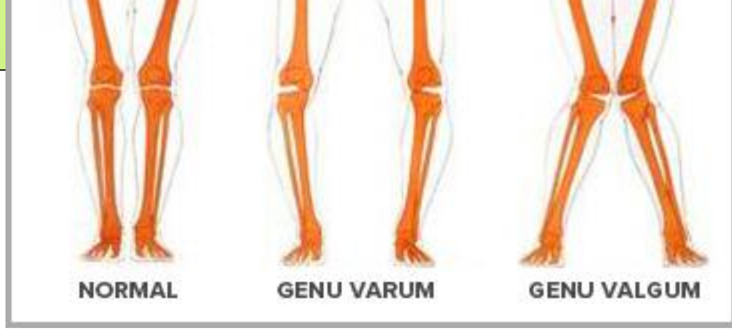
Femoral Anteversion

- HPI: **4-6 year old female** (2X more frequently) with **in-toeing** who often **sits in "W-position"**
- PE: "**kissing kneecaps**", entire leg appears internally rotated
  - Patient lies prone with knee flexed to 90 degrees, allow distal leg to move laterally
- Tx: **Reassurance**, follow-up, **avoid W-sitting**
  - Resolves by 7-8 years
  - <1% require surgery



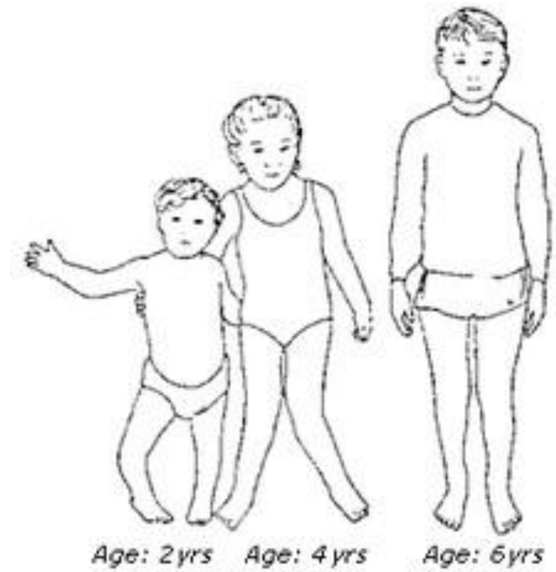
Out-Toeing
External Tibial Torsion

- **Most common** cause
- Often related to **in utero** positioning
- Tibia rotates externally with age (can worsen)
- Tx: **observation & reassurance**
  - Dysfunction, cosmetic concerns → surgery

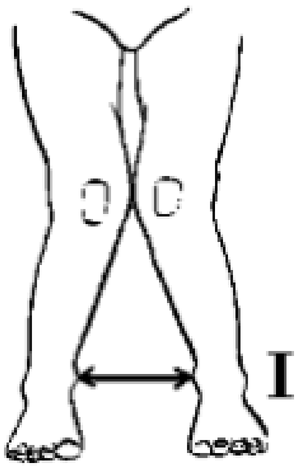


Genu Varum & Genu Valgum

- **Normal variation** in most
  - **Infants-toddlers = genu varum**
  - **4-year olds = genu valgum**
  - Adolescents = straight
- Assess overall height
  - <2 SD may = skeletal dysplasia
- Check diet and activities
  - **Rickets** (bone pain, dental deformities, impaired growth, skeletal deformities)

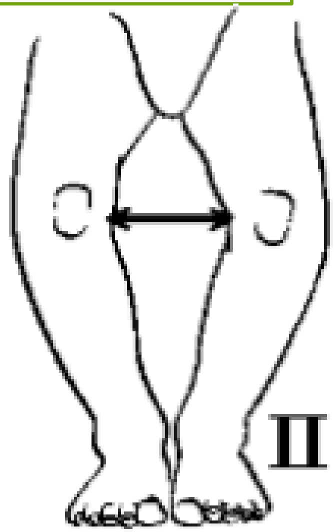


Genu Valgum



- **Most common in 3-4 yo**
- **Resolves by 5-8 yo**
- Dx: **measure distance between the two tibial medial malleoli** with the knees touching
- Tx: **reassurance**
  - Surgery for severe deformities, gait dysfunction, pain

Genu Varum



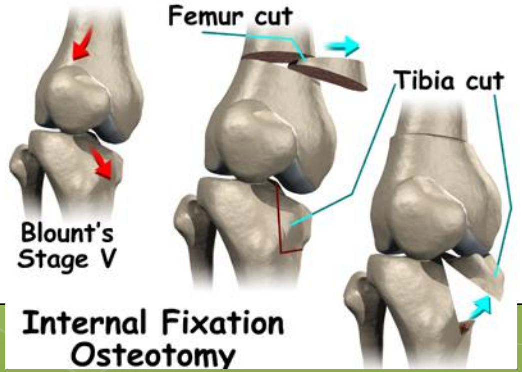
- **Most common > 18 months**
- **Improves by 2-3 years**
- Dx: **measure the distance between the medial femoral condyles** with the medial malleoli touching
- DDx: Need to **rule out tibia vara** (Blount disease)
- Tx: **reassurance**
  - Surgery for severe deformities, gait dysfunction, pain

Tibia Vara (Blount Disease)

- **Abnormal growth of medial aspect of proximal tibial epiphysis**
  - Cause = unknown
- Infantile (1 to 3 years) – most common
  - African Americans, females, obese patients
  - 80% B/L
  - **Painless**
- Juvenile (4 to 10 years)
- Adolescent (>11 years)
  - African Americans, males, markedly obese patients
  - 50% B/L
  - **Painful**

Tibia Vara (Blount Disease)

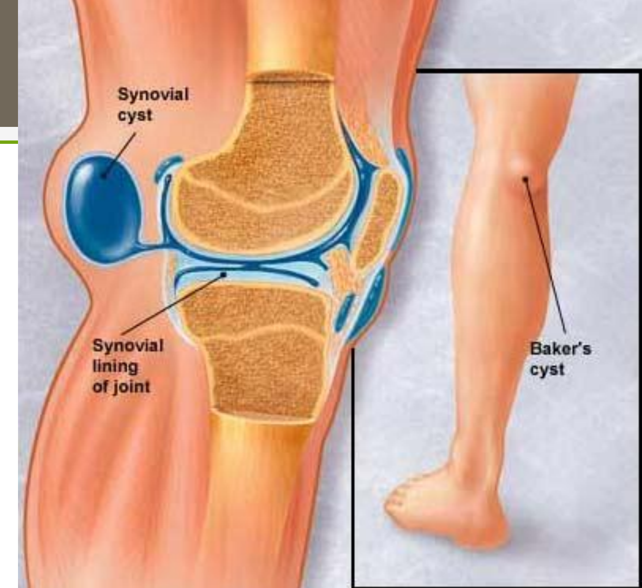
- Dx: **Weight-bearing AP & lateral x-rays of both LE**
  - Fragmentation, **wedging**, **and beak deformities of the proximal medial tibia**
- Tx: **Immediate orthotics** in < 3 yo
  - **Surgery** > 4 yo
    - Proximal tibial valgus osteotomy with fibular diaphyseal osteotomy



The Knee

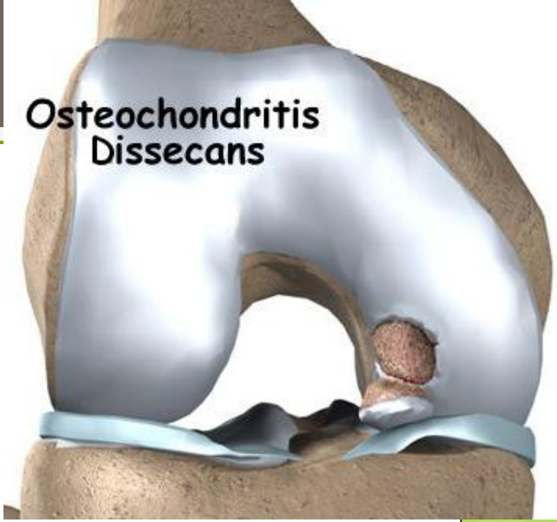
Popliteal Cyst
Osteochondritis Dissecans
Osgood-Schlatter Disease
Patellofemoral Pain Syndrome

Popliteal Cyst (Baker's Cyst)

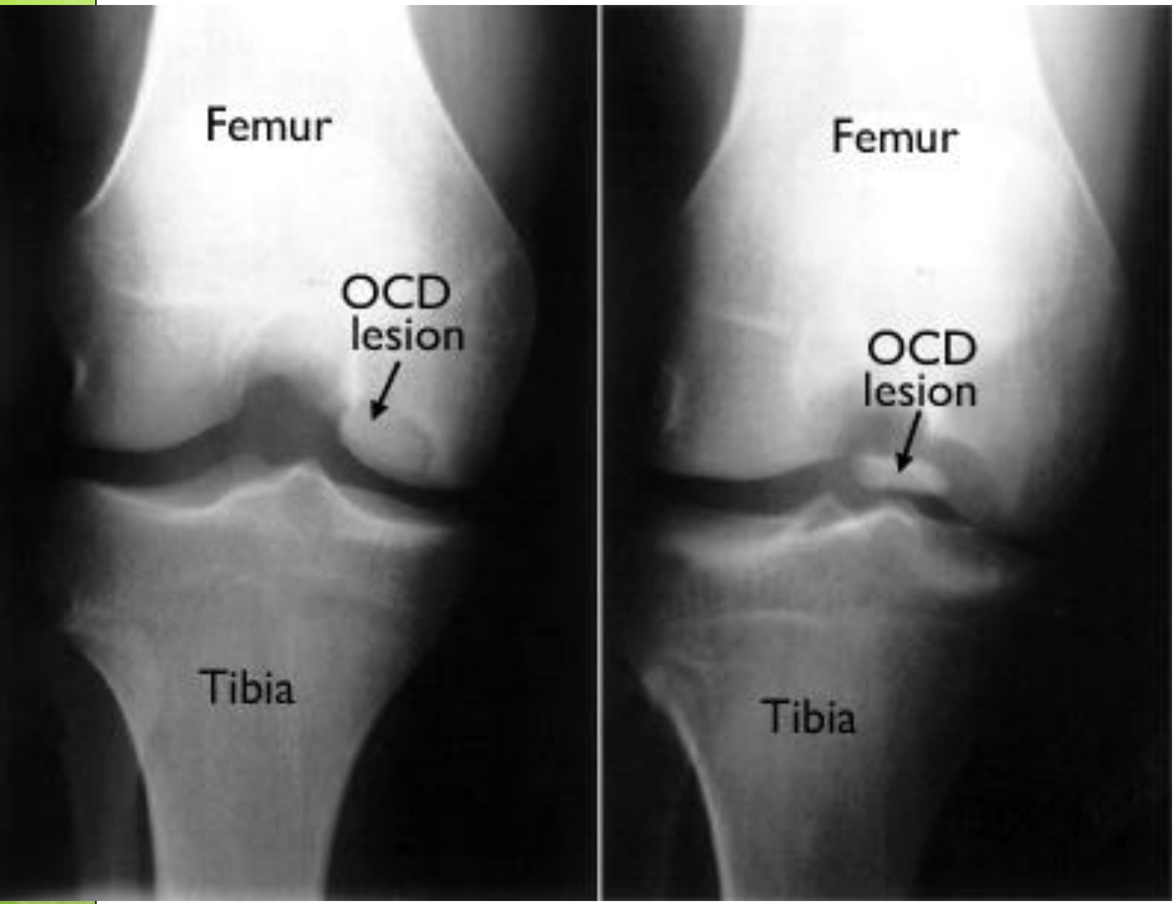


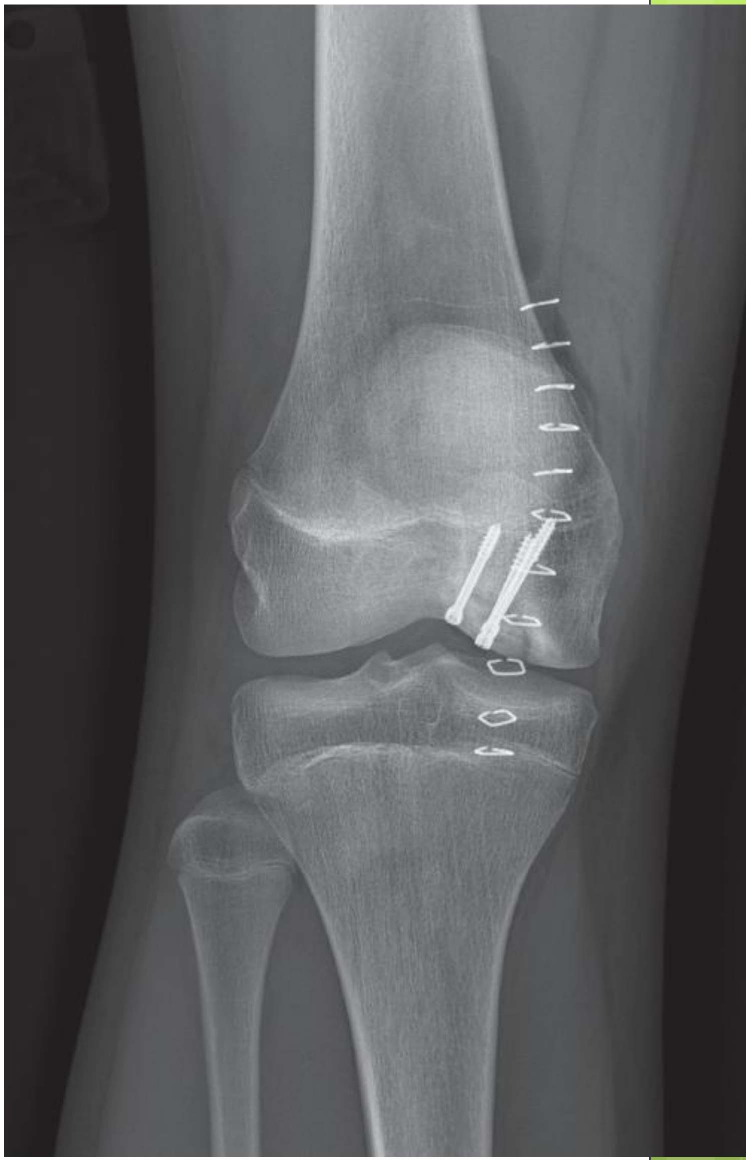
- **Middle childhood** years
- Caused by **distension of the gastrocnemius & semimembranous bursa along the posteromedial aspect** of the knee by synovial fluid
  - In adults they are due to meniscal tears
- In Pediatrics, usually **painless and benign**
- **Resolve spontaneously** (may take years)
- Tx: reassurance
  - Surgery if progressive or limiting mobility

Osteochondritis Dissecans

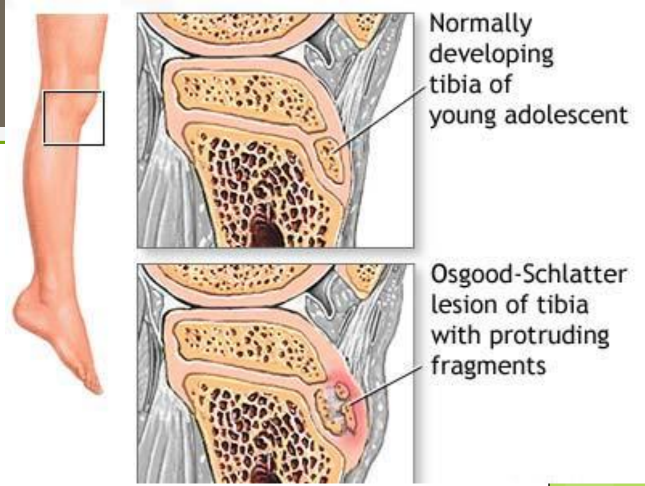


- Most commonly affects **knee**
- Occurs when an **area of bone adjacent to the articular cartilage suffers a vascular insult & separates from the adjacent bone**
  - Lateral aspect of the medial femoral condyle
- HPI: **knee pain and swelling**
- Dx: **MRI**
- Tx: Young patient with **intact articular cartilage** → revascularize and heal with **rest**
  - Older child with **damage to cartilage** → **surgery**


Femur
OCD lesion
Tibia
Femur
OCD lesion
Tibia



Osgood-Schlatter Disease



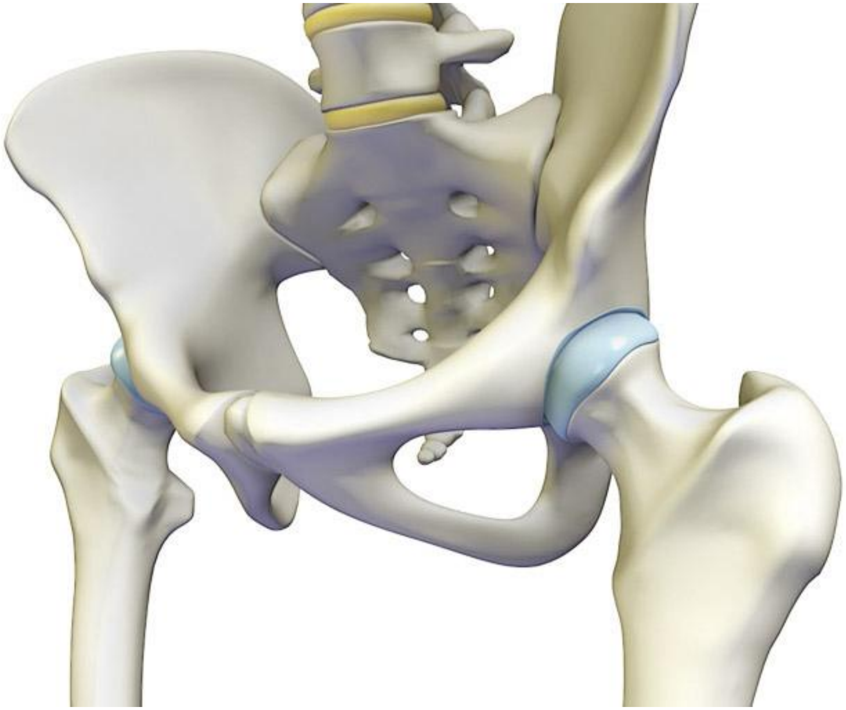
- **Knee pain at the insertion of the patellar tendon on the tibial tubercle**
- Stress from a contracting quadriceps muscle →
- **Microfracture or partial avulsion** fracture in the ossification center
- Usually occurs after a growth spurt
- More common in **boys**
- 11 years ♀, **13-14 years** ♂

Osgood-Schlatter Disease

- HPI: **13 yo M basketball player with knee pain during and after activity**
- PE: **tenderness and edema over tibial tubercle**
- Tx: Rest and activity modification
  - NSAIDs, ice
  - Stretching and LE strengthening
  - Sx typically last 1-2 years
- Complications: **bony enlargement of tibial tubercle & avulsion fracture**

Patellofemoral Pain Syndrome

- Common in **adolescent female athletes**
- HPI: **anterior knee pain**, **worse with activity** & stairs, soreness after sitting in one position for extended time
- **Grinding sensation** under kneecap
- No edema
- PE: **+ tenderness to palpation and compression of patellofemoral joint with knee extended**
  - Weak hip musculature & poor LE flexibility
- Tx: **strengthening of hip girdle** and increased LE flexibility
  - NSAIDs, rest, ice

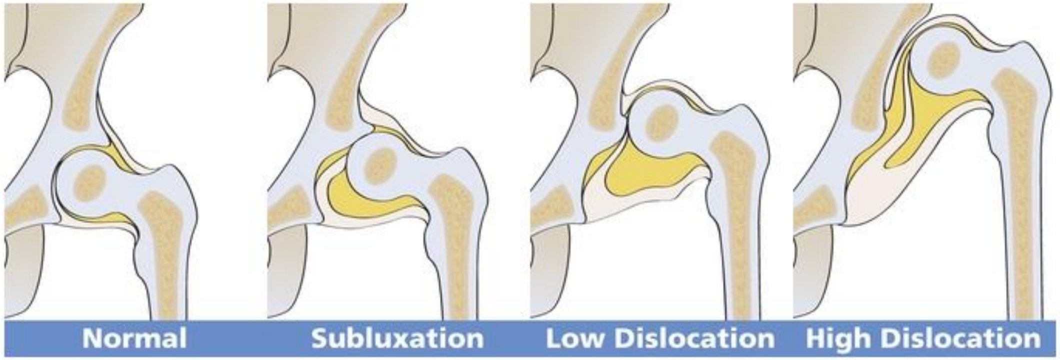


The Hip

Developmental Dysplasia of the Hip
Transient Synovitis
Legg-Calve-Perthes Disease
Slipped Capital Femoral Epiphysis

- Hips at birth may be dislocated or dislocatable
- Newborns have **ligamentous laxity**
  - Spontaneous dislocation & reduction of femoral head
    - Persistence → pathologic changes
      - Flattening of acetabulum
      - Joint capsule tightening
      - Muscle contractures

Severity of Developmental Dysplasia of the Hip (DDH)

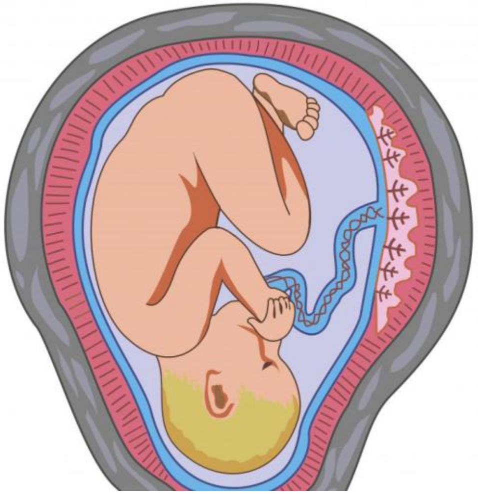


Types of hip dysplasia

Hip dysplasia has a wide range of severity. In some children the ligaments around the hip joint are loose allowing the hip to subluxate. This is when the ball is no longer centered in the socket. Other times the ball is slightly or completely dislocated from the socket.

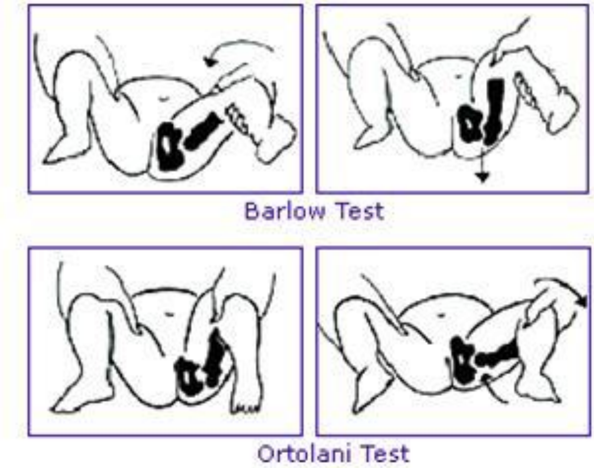
- Risk Factors:

1. **Female** (9:1)
2. **Family history** (20%)
3. **Breech presentation**
4. **Firstborn child** (60%)
5. **Oligohydramnios**



DDH Examination

- Every newborn – toddler has their hips examined at every well child visit
- Asymmetrical thigh and gluteal folds
- **Barlow** – infant supine, hip & knee flexed, adduct thigh with gentle downward pressure (dislocate)
  - < 2-4 months
- **Ortolani** - infant supine, hip & knee flexed, abduct thigh with gentle upward pressure (reduce)
  - < 2-4 months
- **Galeazzi** – infant supine, hip & knee flexed & feet flat on table, examine knee heights for symmetry
  - >3 months



DDH

- Dx: **Physical Exam maneuvers**
  - **<u>Hip US < 4 months</u>**
    - ♀ with + FHx; breech presentation ♀ /♂
  - **<u>Hip x-rays >4 months</u>**
- Tx: **Pavlik Harness** < 6 months of age
  - >6 months = **closed reduction and hip spica cast**
  - If closed reduction fails → open reduction
- Complications: avascular necrosis of femoral head; pressure ulcers

Transient Synovitis (aka Toxic Synovitis)

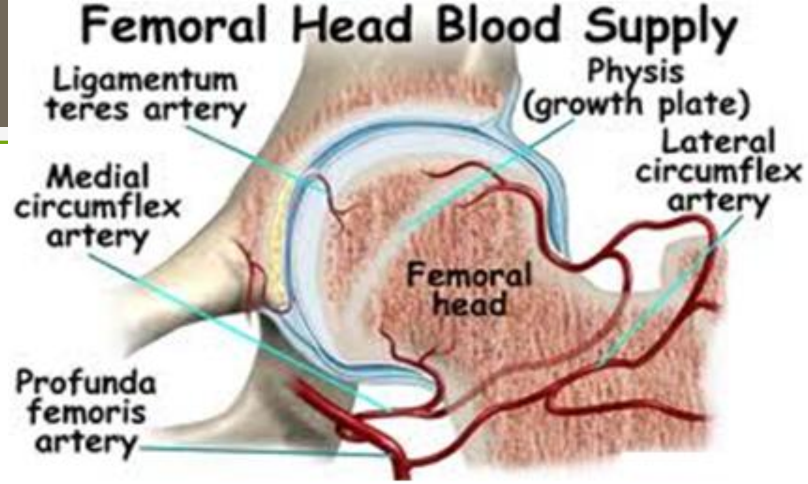
- **Dx of exclusion** (septic arthritis & osteomyelitis)
- HPI: **3-8 yo** (mean = 6 yo) **male** (2X)
  - **Preceding URI** (70%)
  - **Acute onset** of joint pain (hip/knee)
    - **Painful** limp
  - Afebrile
- Dx: **labs, x-rays normal**
- Tx: **Bedrest with minimal weight bearing**
  - NSAIDs

Hip pain often refers to knee
(pt with hip pathology may complain of knee pain)

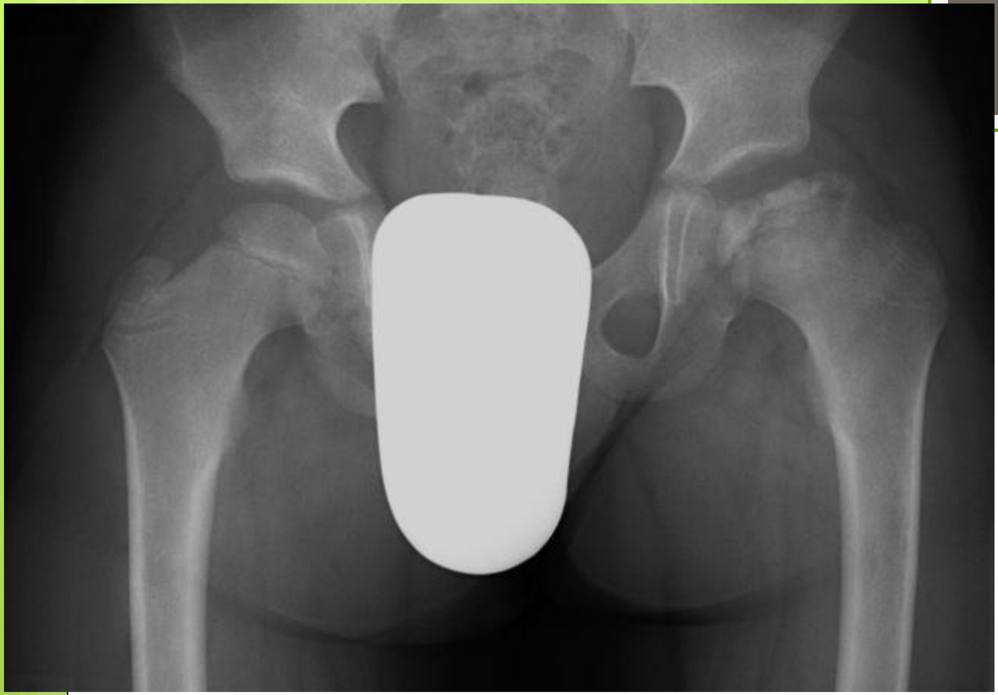
BACTERIAL INFECTION *	TRANSIENT SYNOVITIS
Fever—temperature >38.5° C	Afebrile
Leukocytosis	Normal WBC count
ESR >20 mm/hour	Normal ESR and CRP
Refusal to walk	Painful limp
Hip held in external rotation, abduction, and flexion	Hip held normally

* Septic arthritis, osteomyelitis

Legg-Calve-Perthes Disease



- <u>**Idiopathic avascular necrosis of the femoral head**</u>
- Unknown etiology – possible interruption of blood supply to capital femoral epiphysis
- HPI: **3-12 yo** (mean = 7 yo) ♂ (4-5X) with atraumatic, **painless** limp
  - May have <u>decreased ROM at hip joint</u>, thigh muscle spasm
- Dx: **AP & frog-leg x-rays of hips**
- Tx: Refer to **pediatric orthopedist**
  - Pain control, restore hip motion, prevent complications
    - Femoral head deformity & arthritis
  - Containment (casts, orthoses, surgery)

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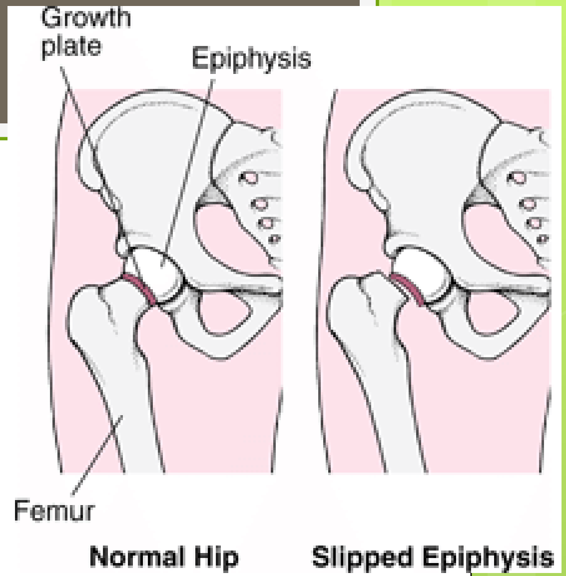
SEJ

Slipped Capital Femoral Epiphysis



- SCFE (pronounced "skiffy")
- **Femoral head is displaced, posteriorly and inferiorly to the femoral neck**
- Requires <u>**immediate**</u> diagnosis and treatment
- HPI: <u>**Obese AA ♂ teenager with acute onset of limp**</u> and hip and/or knee pain

Hip pain often refers to the Knee!!

Slipped Capital Femoral Epiphysis

- PE: may hold their **hip in passive external rotation**; may refuse to bear weight
- Dx: send **<u>via wheelchair</u>** for **AP and frog-leg** x-rays of hips
- AP radiograph: The <u>Klein line</u> is drawn straight up the superior aspect of the femoral neck. This should intersect the epiphysis. If not, then it is likely an SCFE

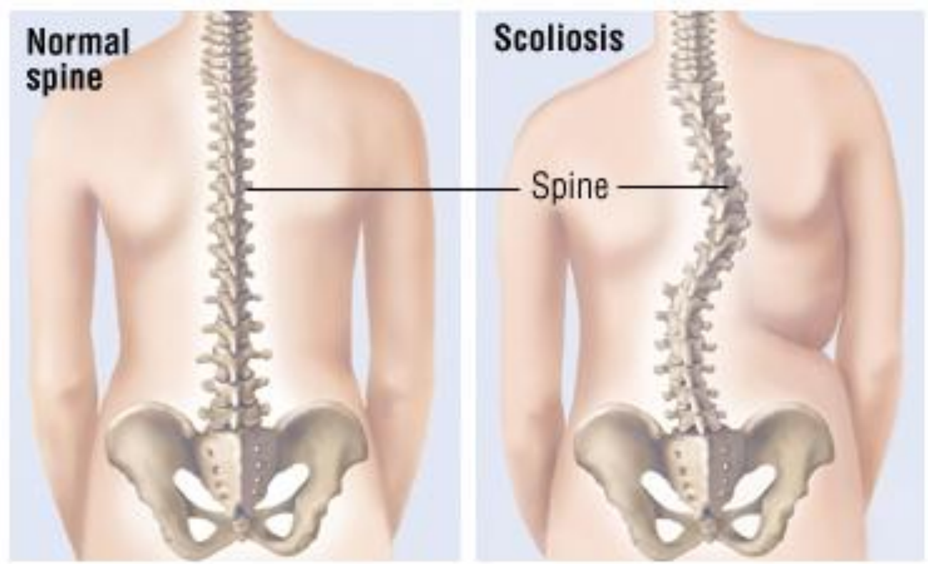
SCFE

- Tx: **Emergent surgery** (internal fixation)
- Complications: increased severity of SCFE; early degenerative arthritis; avascular necrosis

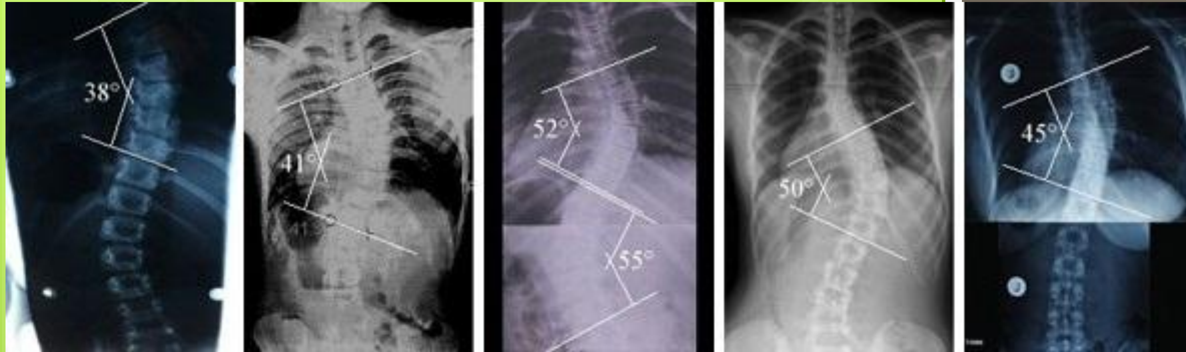
The Spine

Scoliosis
Kyphosis
Torticollis
Spondylolysis
Spondylolisthesis
Diskitis

Scoliosis



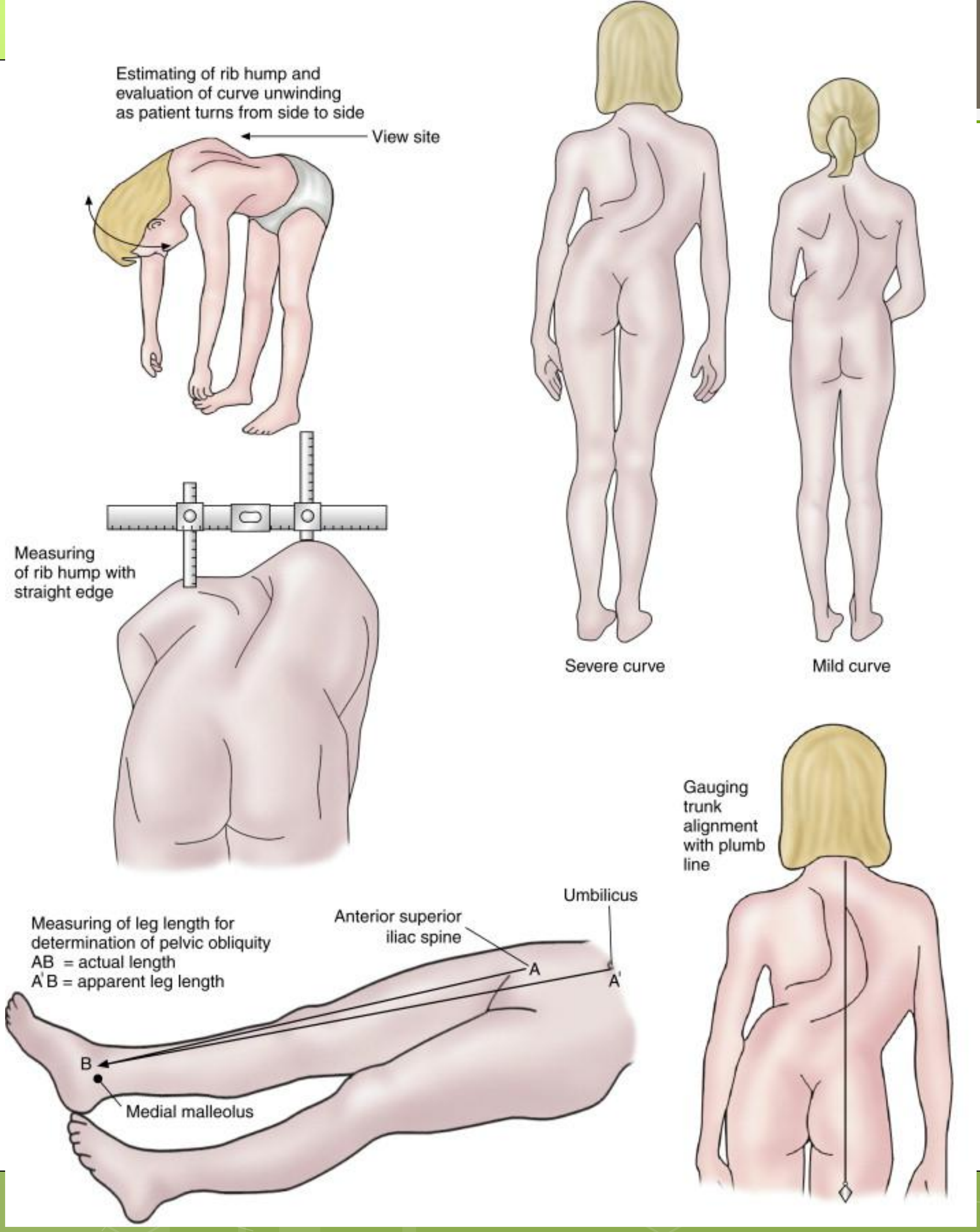
- Alterations in normal spinal alignment that occur in the AP plane = scoliosis
  - **Idiopathic** → Most common!
  - Congenital → Abnormalities of vertebral formation in the 1st trimester
  - Neuromuscular → Progressive: CP, muscular dystrophy, spina bifida
  - Compensatory → Adolescents with leg-length discrepancy



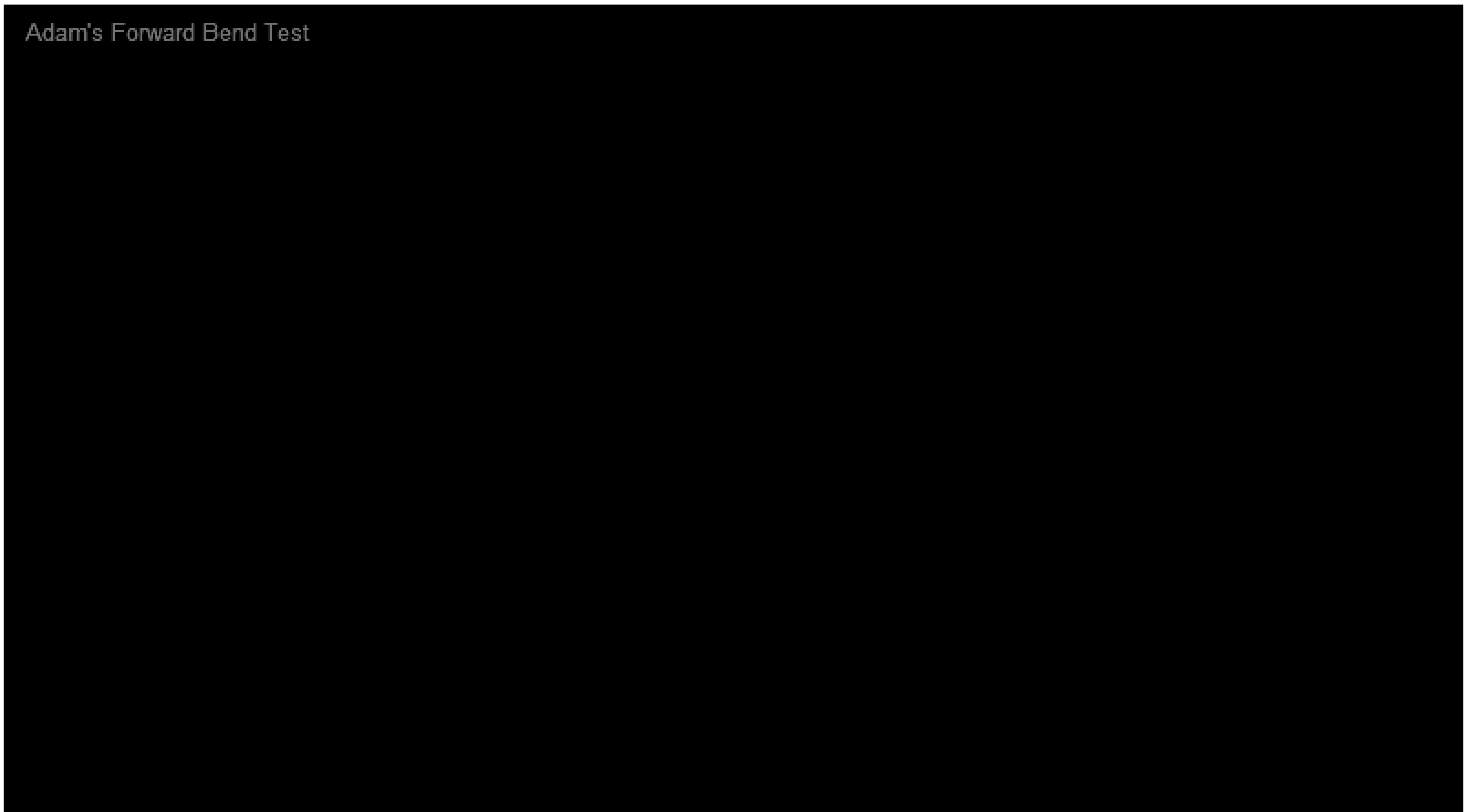
Scoliosis

- Dx: **examine back of patient from behind**
  - Check for **pelvic asymmetry**
  - Check for **cutaneous lesions**
  - Palpate for **tenderness**
  - **Adam's forward bend test**
- If any of the above are +:
  - **Scoliosis Series** = PA & lateral x-rays of the entire spine
    - Measure **Cobb angle**

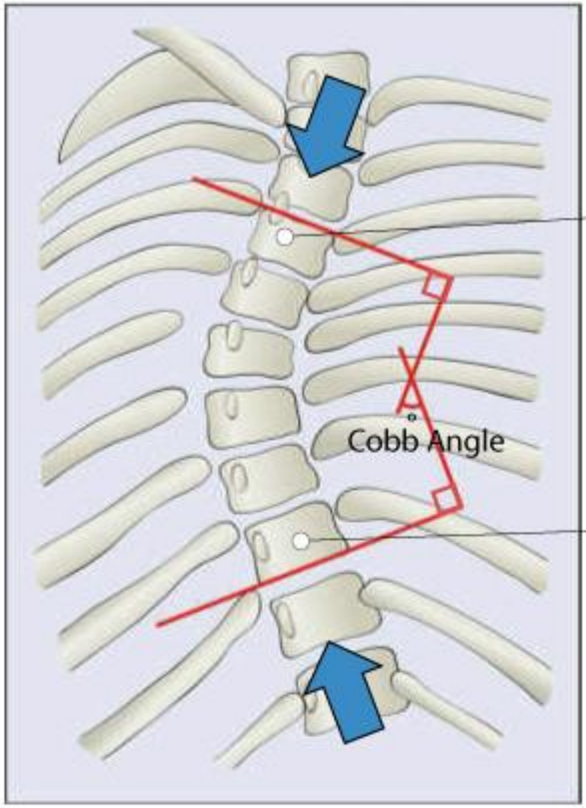
Stand with feet together


Estimating of rib hump and
evaluation of curve unwinding
as patient turns from side to side
View site
Measuring
of rib hump with
straight edge
Severe curve
Mild curve
Gauging
trunk
alignment
with plumb
line
Umbilicus
Anterior superior
iliac spine
Measuring of leg length for
determination of pelvic obliquity
AB = actual length
A'B = apparent leg length
A
A'
B
Medial malleolus

Adam's Forward Bend Test



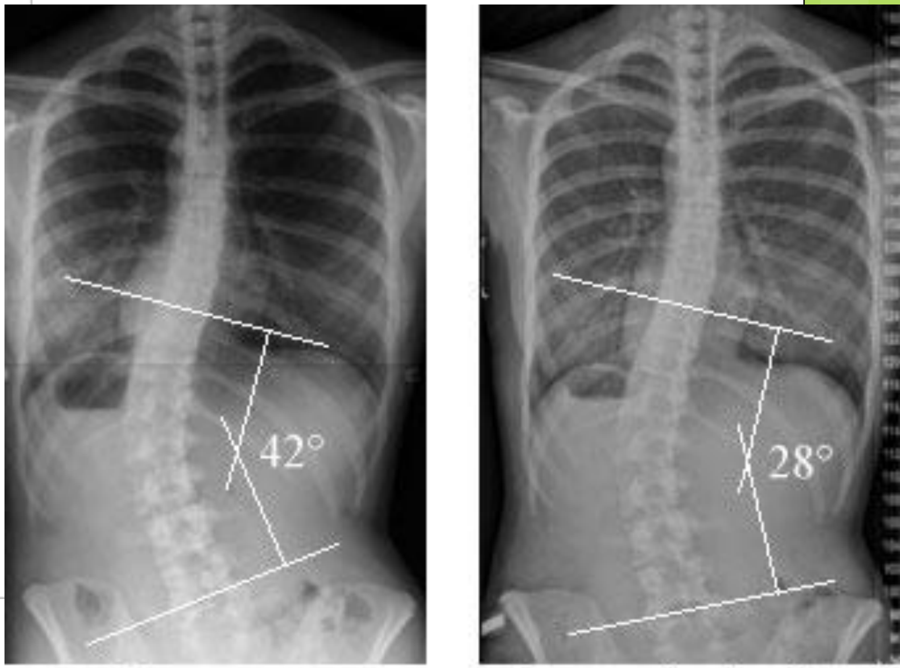
MEASURING THE COBB ANGLE



From the top, the most displaced vertebrae

From the bottom, the most displaced vertebrae

Source: e-radiography.net and core concepts



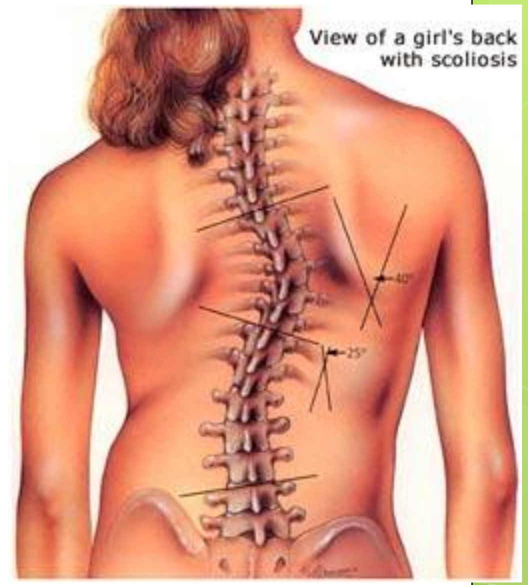
Before Treatment

July 2009
16-years-old
42°

After 4-weeks of ScolioGold

January 2010
17-years-old
28°

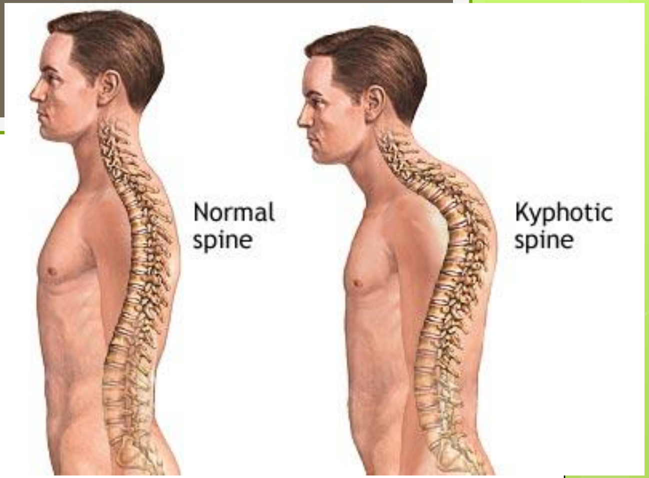
Idiopathic Scoliosis



- Occurs in healthy, neurologically normal children
- Family history (20%)
- ♀>♂ & more progressive in ♀
  - Infantile (birth-3 years)
  - Juvenile (4-10 years)
  - Adolescent (>11 years)
- **R thoracic curvature** (most common) → 80%
- <u>Painless</u> (70%)
- <u>L-sided curve → MRI</u> for eval of intraspinal pathology (tumor or syrinx)

Idiopathic Scoliosis Treatment

- **OMT!**
- Typically **< 25°** → **observation**
- **20°-50°** (in skeletally immature patients) → **bracing**
- **>50°** (or with any **pain**) → **surgery**



Kyphosis

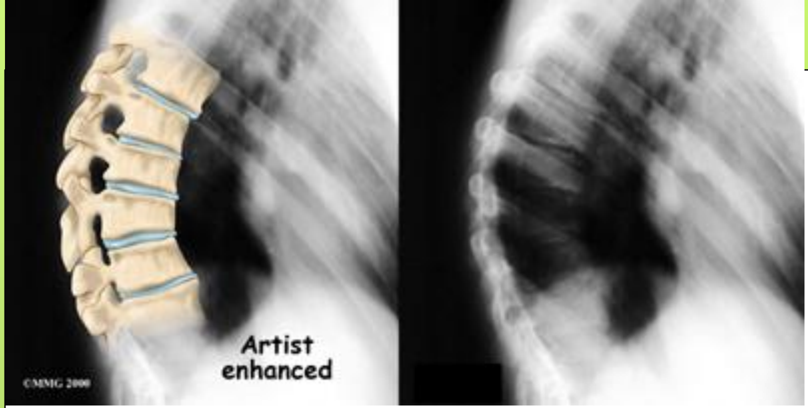
- **Increased angulation** of the thoracic or thoracolumbar spine in the sagittal plane
- **Postural**
- **Structural**
- **Congenital**

Secondary to poor posture

Voluntarily corrects

AKA Scheuermann Kyphosis

Failure of formation of all/part of the vertebral bodies



Scheuermann Kyphosis

- HPI: ♀=♂
- ~50% have back pain
- Kyphosis **does not correct with standing/lying**
- Dx: Forward bend = **abrupt angulation**
  - Smooth contour in postural roundback
- X-ray = **narrow disk space, anterior wedging**
- Tx: **OMT**, **bracing**, **surgery**

Torticollis

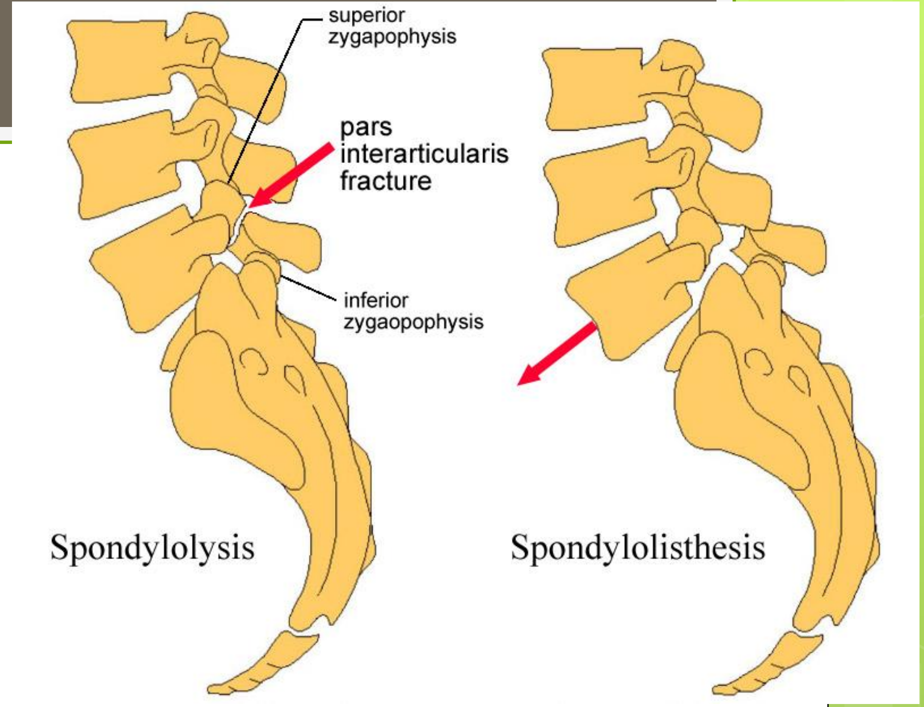
- HPI: **Newborn with a head tilt**
- **Muscular torticollis**
  - 2/2 **shortened sternocleidomastoid** muscle
  - Birth trauma vs in utero positioning
  - May have **palpable mass**/swelling in SCM
- **Acquired torticollis**
  - **Upper cervical spine abnormality** vs. **CNS pathology** (mass lesion)

Exercise 1. Child undergoing exercise with face looking toward shoulder.

Torticollis

- Dx: **clinical**
  - +/- AP & lateral cervical spine films
  - MRI head/neck for persistent pain, neuro symptoms, persistent deformity
- Tx: **OMT!** (ME)
  - **Stretching**
  - **Surgery with post-op OMT/PT**

Exercise 2. Child undergoing exercise with ear tilting toward shoulder.

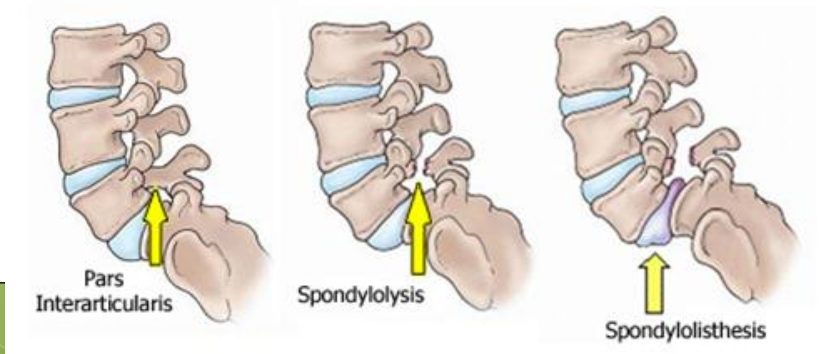


Spondylolysis

- Defect in the **pars interarticularis**
- Most common at **L5 or L4**

Spondylolysthesis

- **Bilateral defects with anterior slippage of the superior vertebra on the inferior vertebra**
- Most common at L5 on S1

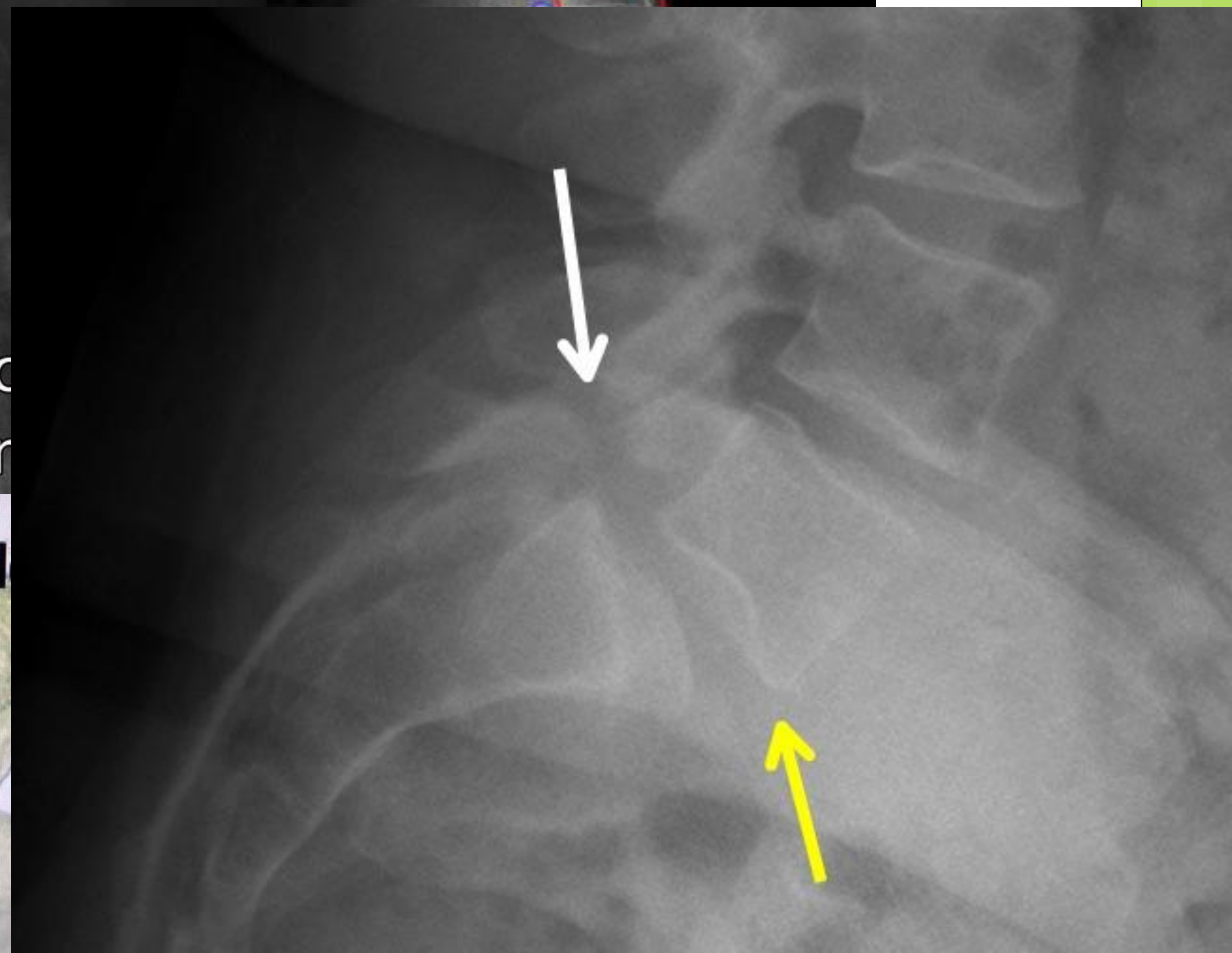


Spondylolysis & Spondylolisthesis

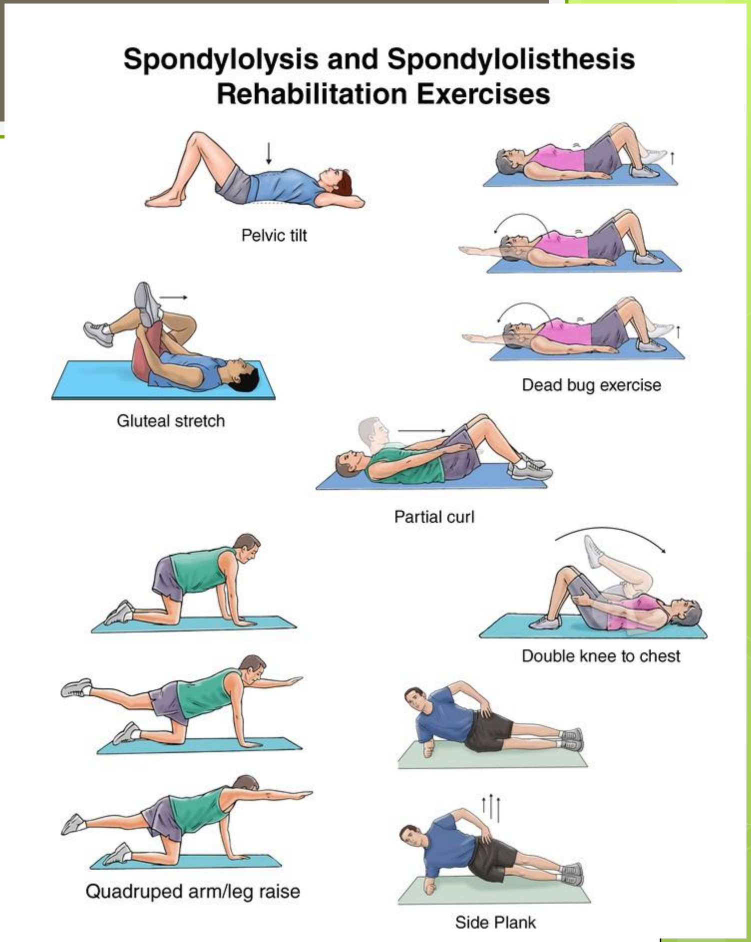
- HPI: **Adolescent athlete (repetitive back extension**: gymnastics, diving, football, soccer, basketball) with insidious onset of **LBP X 2 weeks**
  - **Pain worsens with activity**, may radiate to buttocks, improves with rest
- PE: **+tenderness to palpation** of affected vertebrae
- **Palpable step-off at the LS area** (spondylolisthesis)
- Need to do a detailed neuro exam
- Dx: AP, lateral & oblique spine x-rays; CT

The “Scotty Dog”
Oblique X-Ray c
the Lumbar Spir

Spondyl



Spondylolysis & Spondylolisthesis



- Tx: Dependent on severity
- **Activity Restriction**
- **OMT/PT**: improve LE flexibility, increase core strength & spinal stability
  - Begin with flexion exercises
- **Surgery** (if pain is severe, ROM greatly restricted, or slippage is progressing)
  - Spinal fusion

Diskitis

- **Intervertebral disk space infection** that does not cause associated vertebral osteomyelitis
  - ***Staphylococcus aureus***
- HPI: **<6 yo child with back pain**, abdominal pain, irritability, <u>refusal to walk/sit</u>
  - +/- fever
- Dx: **Refusal to flex lumbar spine**, paravertebral muscular spasm
- <u>AP, lateral, oblique spine x-rays</u> → narrow disk space
- **MRI** (diskitis vs. vertebral osteomyelitis)
- Tx: **IV antibiotics X 1-2 weeks + 4 weeks PO**

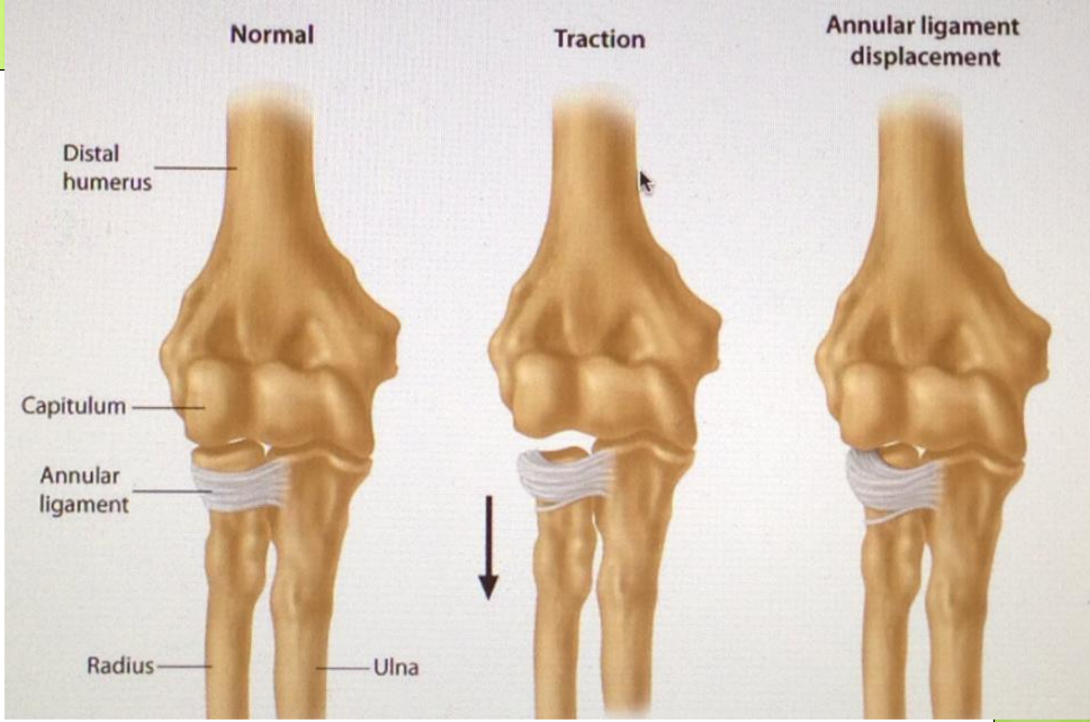
Upper Extremity

Radial Head Subluxation

Upper Extremity

Name	HPI	Dx/Tx
Sprengel Deformity	**Congenital elevation of the scapula**; usually unilateral	Mild = reassurance Moderate-severe = surgery
Brachial Plexus Injuries	Either **newborn** (obstetric brachial plexus palsy), or **Athletic injury** (lateral flexion of neck away from UE; direct impact to BP; neck extension & rotation toward UE)	Athletic: Usually unilateral and resolve within 15 minutes Must assess cervical spine for serious injury
Proximal humeral epiphysiolysis *(aka Little Leaguer's shoulder)*	**9-14 yo baseball pitcher** with pain during/after throwing	X-rays, rest, OMT & strengthening exercises

Radial Head Subluxation (Nursemaid's Elbow)



- Caused by a **quick pull on the extended elbow**
- HPI: **Infant/Toddler with hand held in pronated position**; refusal to move hand or elbow
- Dx: **clinical** (x-ray if suspect fx)
- Tx: **Supinate hand while applying pressure to radial head**
- High rate of recurrence


Nursemaid's Elbow


Be Careful When U Hold
or Carry Your Child Hand
www.facebook.com/learningpetals
Dislocación
de la cabeza
del radio
Radio
Cúbito
Húmero
No levantar al niño
tomándolo de las mano
o de las muñecas
Sólo el médico debe
acomodar el ligamento
que sale de su lugar
Luego de la
intervención
médica,
la recuperación
es inmediata
Don't Forget To Share

Reference

- Kliegman RM. *Nelson: Essentials of Pediatrics*. 7th Edition. Saunders; 2014
  - Chapters 199-203
  - *Text available online via TUCOM library (Call# RJ45)*

- https://www.youtube.com/v/imhI6PLtGLc = Barlow & Ortolani Tests
- https://www.youtube.com/v/n__zHzbdsEI = Adam's Forward Bend Test
-
- https://www.youtube.com/v/tJb5rGOFiTY = Reducing a Nursemaid's Elbow